



Investigating the role of surface micro/nano structure in cell adhesion behavior of superhydrophobic polypropylene/nanosilica surfaces



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ARTICLE INFO

Article history:

Received 13 December 2014

Received in revised form 20 January 2015

Accepted 29 January 2015

Available online 7 February 2015

Keywords:

Superhydrophobic

Cell adhesion

Micro/nano structure

Nanosilica

Non-solvent

ABSTRACT

The main aim of the current study was to investigate the effects of different topographical features on the biological performance of polypropylene (PP)/silica coatings. To this end, a novel method including combined use of nanoparticles and non-solvent was used for preparation of superhydrophobic PP coatings. The proposed method led to a much more homogeneous appearance with a better adhesion to the glass substrate. Moreover, a notable reduction was observed in the required contents of nanoparticles (100–20 wt% with respect to the polymer) and non-solvent (35.5–9 vol%) for achieving superhydrophobicity. Surface composition and morphology of the coatings were also investigated via X-ray photoelectron spectroscopy and scanning electron microscopy. Based on both qualitative and quantitative evaluations, it was found that the superhydrophobic coatings with only nano-scale roughness strongly prevented adhesion and proliferation of 4T1 mouse mammary tumor cells as compared to the superhydrophobic surfaces with micro-scale structure. Such results demonstrate that the cell behavior could be controlled onto the polymer and nanocomposite-based surfaces via tuning the surface micro/nano structure.

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1. Introduction

Superhydrophobic surfaces have attracted considerable interests over the past decade due to their promising potential applications in numerous fields [1–6]. Such attention can be attributed to the unique wetting properties of these surfaces, for instance extreme water-repellent, self-cleaning, anti-fouling, anti-icing, anti-sticking and anti-contamination. Based on the general definition, a water droplet on superhydrophobic surfaces keeps almost its spherical shape with a water contact angle higher than 150° and sliding angle lower than 10°; therefore, it can easily roll off when the surfaces are slightly tilted [7]. Superhydrophobicity like other numerous phenomena has been inspired from nature. Many plant leaves and insects for example lotus leaves and butterfly wings have the superhydrophobic property. Although the surface

chemical composition is an important factor for obtaining superhydrophobicity, its roughness is also vital. The maximum water contact angle that can be achieved by a smooth non-porous surface of closest hexagonal packed $-CF_3$ groups is 120° which has been obtained in Teflon® surfaces [8]. Therefore, surfaces having contact angles higher than 150° can only be attained through inducing roughness and porosity [9].

Generally, there exist two approaches for preparing superhydrophobic surfaces, that is, to make a rough surface from a low surface energy material or modify a rough surface with a material of low surface energy [10]. Up to now, numerous methods have been developed to fabricate superhydrophobic surfaces, for example sol–gel process, template-based techniques, chemical etching, self-assembly procedure, phase separation and others [11–13]. So far, many of the used methods are either expensive or require rigorous preparation conditions which limit their applications.

Polypropylene (PP) is an intrinsically hydrophobic material, due to having no polar groups in its chemical composition, with a melting temperature about 170 °C, and shows various morphologies

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based on the employed processing conditions [14]. Engineering materials such as PP, polyvinyl chloride, polyethylene, polymethylmethacrylate with superhydrophobic properties are the most likely candidates for applications in daily life. Due to PP's balanced physical and chemical properties, nontoxicity, and low cost, it has been widely used in many areas such as home appliances, automotive, and package films [14]. Therefore, the surface wettability is critical for its practical applications [15]. In addition, PP-based nanocomposites have also been under intense investigations within the last two decades [14,16,17].

Several authors have studied superhydrophobic PP surfaces within the recent years. They have used very expensive and complex methods to fabricate superhydrophobic surfaces. Rasilainen et al. [18] prepared superhydrophobic PP surfaces by injection molding. They also fabricated superhydrophobic hierarchical-structured PP surfaces using template method [9]. In 2003, Erbil et al. [19] reported a simple approach for fabricating superhydrophobic surfaces with PP via controlling the phase separation process by adding proper amounts of non-solvent. Since then, numerous papers have been published based on this method for a wide range of polymers and nanocomposites. Hou et al. [20] fabricated superhydrophobic PP surfaces by using a proper combination of non-solvents via phase separation method. In another work by our group, superhydrophobic surfaces were prepared using PP and the corresponding nanocomposites containing zinc oxide nanoparticles [21].

In the field of biomedicine and tissue engineering, investigation on the interactions between cells/proteins and different surfaces is of crucial importance from both theoretical and practical viewpoints [22,23]. Most known mammalian cells have the tendency to adhere onto a surface in order to conduct normal metabolism, proliferation and differentiation [24]. Apart from physiological activity of cells, it is known that cell adhesion is highly governed by surface properties including surface energy, roughness, and chemical composition [25]. Such complex interrelationships are the possible reasons for the contradictory reported behaviors. For example, superhydrophobic surfaces were reported to be extremely cell repellent [26,27] whereas it was shown, in another work, that cells not only adhere but also proliferate on superhydrophobic surfaces [28]. Therefore, in this study, two types of superhydrophobic surfaces with different surface topographical features were prepared in order to determine their corresponding influence on the cell adhesion behavior. To this end, a combined use of silica nanoparticles and a proper solvent/non-solvent system is suggested to significantly reduce the extent of needed nanoparticles and non-solvent to achieve superhydrophobicity. The complete coverage of polymer surface by nanoparticles is not necessary in this method, and instead, formation of a unique surface topography together with partial migration of nanoparticles onto the top layer could be regarded as the main criteria for achieving ultra water repellent surfaces. For the first time, such method has been presented by our group, in which the used solvent/non-solvent system was THF/ethanol [2]; however, the current system is p-xylene/methyl ethyl ketone (MEK). Moreover, as mentioned earlier, cell culture was conducted for the as-prepared control and superhydrophobic surfaces to determine the effect of surface topography on the cell adhesion property.

2. Material and methods

2.1. Materials

PP (SEETEC H5300, density = 0.9 g/cm³) was purchased from LG Corporation with melt flow rate of 3.5 g/10 min which was measured under a load of total mass of 2.16 kg at a temperature of

230 °C. The hydrophobic fumed silica, used in this study, is a commercial product (Aerosil R805) which was purchased from Evonic Co. (Germany) and used as received. Aerosil R805 has a specific surface area of $150 \pm 25 \text{ m}^2 \text{ g}^{-1}$ and primary particle size of 12 nm, and is produced by treating SiO₂ with Octylsilane (C₈H₁₇SiH₃). Methyl ethyl ketone (MEK) and Xylene were supplied by Merck and used as received.

2.2. Preparation of PP coatings

All the superhydrophobic surfaces were fabricated through typical solution casting technique. Regarding singular use of non-solvent, 200 mg PP granules were dissolved completely in 10 mL p-xylene at 120 °C. After preparing a homogeneous solution, different contents of MEK, as a non-solvent, (0, 1, 2, 3, 4, 5, 6, 7 and 8 mL) were added drop-wise into PP solution under stirring and then dispersed for 2 min prior to drop casting on the pre-cleaned glass slides. Regarding singular use of silica nanoparticles, different amounts of nanosilica (50, 100, 150 mg) were first vigorously stirred in 10 mL PP solution, and after that, mixed in an ultrasonic bath for 1 h prior to drop-casting onto the substrates. Finally, regarding combined use of nanosilica and non-solvent, different amounts of nanosilica (10, 20 mg) were dispersed in 10 mL p-xylene followed by addition of various contents of MEK (1, 2 mL) into PP/nanosilica solution under stirring, after that, the solution was mixed for another 2 min prior to drop casting on the pre-cleaned glass slides. For simplicity, the prepared non-solvent-based samples are named MEK1, MEK2, MEK3, MEK4, MEK5, MEK6 and MEK7 in which the number accounts for the volume of added MEK into 10 mL p-xylene. Moreover, the samples containing different amounts of nanosilica are named Si50, Si100 and Si150, and the samples containing both nanosilica and MEK are named MEK1Si10, MEK1Si20 and MEK2Si10.

2.3. Characterization

A video-based contact angle measurement system (OCA 15, DataPhysics Instruments GmbH, Filderstadt, Germany) was employed to determine the water contact angle values of the samples. The water contact angle measurements of each sample was conducted at least three times across the sample surface using the sessile drop method by dispensing 4 μL drops of deionized water (surface tension $\gamma_{lv} = 72.8 \text{ mN m}^{-1}$ at 25 °C) on the samples surfaces. All water contact angle values were measured under ambient laboratory conditions at 25 °C.

Morphologies of the coating surfaces were evaluated on a digital scanning electron microscope (KYKY-EM3200, KYKY Technology Development, Beijing, China) operated at 25 kV. To avoid electric charging all samples were plated with gold particles.

X-ray photoelectron spectroscopy (XPS) analysis was performed by using XPS spectroscopy with a monochromatic AlK α X-ray source (1486.6 eV photons), operated at 180 W (12 kV and 15 mA) and under ultra-high vacuum conditions.

2.4. Cell-culture experiments

The 4T1 mouse mammary tumor cell line (estrogen receptor (ER)/progesterone receptor (PR)-negative) (National cell bank of Iran, Pasteur institute, Tehran, Iran) was cultured in RPMI 1640 medium (Gibco BRL Inc., Grand Island, NY, USA), supplemented with adding 10% heat-inactivated fetal bovine serum, 1% penicillin/streptomycin, 1 mM pyruvate and 2 mM L-glutamine, and maintained in humidified incubator at 37 °C in a 5% CO₂ atmosphere. Tumor cell suspensions of greater than 90% viability were prepared from subconfluent cultures with 0.25% trypsin (Invitrogen, CA, USA) and 0.02% EDTA.

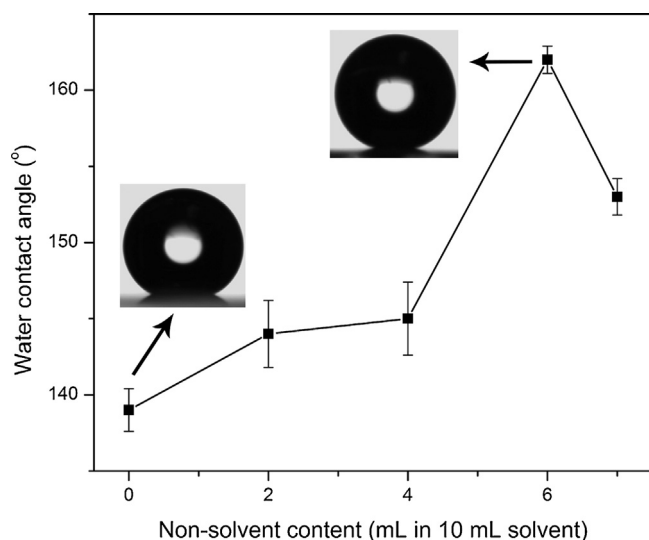


Fig. 1. Water contact angle values in terms of MEK content (non-solvent content in 10 mL solvent).

To prepare samples for SEM observation, at the end of incubation (24 h), samples were washed with PBS to remove unattached cells. The adhered cells were then fixed by glutaraldehyde solution (2.5%) for 10 min and dehydrated through a series of ethanol/water mixtures (10, 30, 50, 70, 80, 90, and 100%) each for 10 min. Dehydration was followed by air drying overnight.

2.5. MTT assay

Cell proliferation was quantified by means of MTT assay, a standard calorimetric assay, in which yellow MTT (3-(4,5-dimethylthiazol-2-yl)-2,5 diphenyltetrazolium bromide, a tetrazole) is reduced to purple formazan. Briefly, the coating samples were washed with phosphate buffer, transferred into new plates containing 5:1 ratio of media and MTT solution (5 mg/mL in phosphate buffer), respectively, and incubated for 2 h at 37 °C. After removing the culture media, 0.5 mL of dimethyl sulfoxide (DMSO) was added to dissolve the insoluble purple formazan product into a colored solution. The absorbance of the supernatant was recorded by a microplate reader at 540 nm. Cell population was determined through a standard curve which was established through a certain number of cells counted by a Neubauer-counting chamber. This test was done according to the procedure presented by Zund et al. who have developed a linear correlation between the cell number and the MTT absorbency at 550 nm [29].

3. Results and discussion

3.1. Wettability, surface morphology and composition

3.1.1. Singular use of non-solvent

Isotactic polypropylene (i-PP) is a hydrophobic polymer with melting temperature of about 170 °C. The water contact angle of the molten i-PP between two glass slides is ~104°. To evaluate the hydrophobic properties, water contact angles and sliding angles were determined for the samples with varied amounts of non-solvent. Variation of water contact angle in terms of non-solvent content together with the corresponding water drop profiles are illustrated in Fig. 1. Without non-solvent, PP can form a surface with water contact angle of 139°. Upon addition of MEK to PP solution, the water contact angle values are increased; however, a decreasing trend is seen when the MEK content exceeds 6 mL. It is evident that low contents of non-solvent into the PP solution results in an

insignificant enhancement in the water contact angle values. It is worthy to note that a considerable inhomogeneity was observed in the sliding angle (water drop stickiness) for these two samples, which could be seen in Fig. 1 (larger error bars). An extremely water repellent behavior (low sliding angle) was observed at the center of the coatings whereas a rather high level of drop stickiness (sliding angle ~50°) was detectable at the surface corners. However, such discrepancy in wettability behavior was disappeared once the MEK content was increased to 6 mL since the whole surface exhibited the same superhydrophobic and rolling-off behavior, and according to the results reported in Fig. 1, the maximum water contact angle value (162°) was obtained for this sample on which the water droplet could easily roll off (SA < 3°). This notable change upon addition of non-solvent content might lead one to postulate that the non-solvent content was not sufficient in the case of MEK2 and MEK4 to establish interactions with all PP macromolecules existing on the coating surface resulting in the observed inhomogeneity in wettability behavior.

To ensure that MEK6 was the optimum sample, a slightly higher content of non-solvent was also prepared, and it was found that addition of higher amounts of non-solvent could be detrimental on the superhydrophobicity of the coating; albeit this sample (MEK7) could also be regarded as superhydrophobic but with a lower water contact angle value.

To further analyze the observed wettability properties, SEM was employed and the results are displayed for two samples, made from different contents of non-solvent (Fig. 2a and b). It is obvious that surface morphology of MEK6 is much more uniform and interconnected than that of MEK4. One could infer that such morphological difference is responsible for the observed difference in water contact angle values. Moreover, in the case of MEK6, a completely uniform morphology comprised of micro-spheres (diameter range of 1–2 μm) can be detected. Such ordered and ultra-fine structure has led to a superhydrophobic surface exhibiting self-cleaning behavior. Another important factor is the much higher homogeneity in the sliding angle values throughout the whole surface. It seems that the content of non-solvent has been increased enough for confronting all the PP macromolecules existing on the surface leading to a much more uniform morphology.

Another interesting observation from Fig. 2a and b is the presence of a high level of porosity, and thus, voids on the coating surface, in which a large amount of air could be enclosed. On such porous superhydrophobic surface, the fraction of air enclosed within the pores is mainly in touch with the water droplet instead of the solid surface according to the Cassie–Baxter model which will be discussed in the following.

Two models have been generally used for describing the extreme wettability behaviors. The first is Wenzel model [30], according to which, water contact angle on a rough surface made by material with intrinsic water contact angle higher than 90° should be enhanced toward superhydrophobicity values while a rough surface with an intrinsic water contact angle lower than 90° tends to superhydrophilicity. In this model, the water droplets penetrate into the pores and cannot roll off the surface easily. In contrary, based on Cassie–Baxter model [31], since the air enclosed between the solid surface and the water droplet can minimize the contact area, the hydrophobicity of a surface will be increased by increasing the surface roughness. On such a patchwork of solid and air, water droplets could not penetrate into the pores and they may only contact with the tips of the polymer micro/nano features. In Cassie–Baxter wetting mode, water contact angle on the structured nanocomposite surface (θ_{SH}) can be expressed as:

$$\cos \theta_{SH} = f_1 \cos \theta_0 - f_2 \quad (1)$$

where θ_{SH} and θ_0 are static water contact angles on the superhydrophobic surface and on a smooth and untextured surface of the

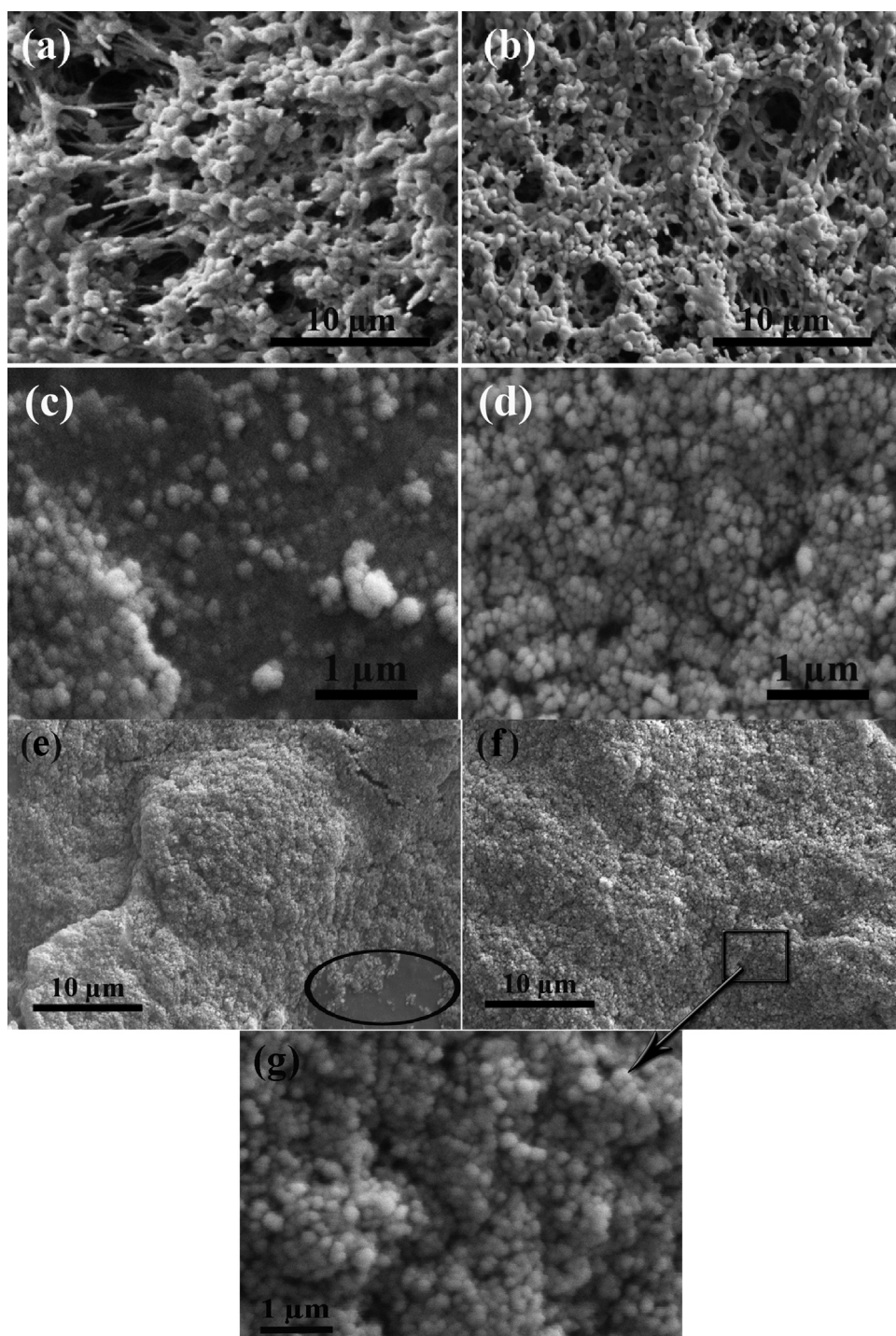


Fig. 2. SEM images regarding singular use of non-solvent: (a) MEK4, (b) MEK6; and regarding singular use of nanoparticles: (c) Si50 and (d) Si100; and regarding combined use of nanoparticles and non-solvent: (e) MEK2Si10, (f) MEK2Si20 and (g) higher magnification of (f).

same composition, respectively, and f_1 and f_2 are the area fractions of the polymer surface and air in contact with water, respectively (i.e., $f_1 + f_2 = 1$). It is evident from Eq. (1) that increasing f_2 increases θ_{SH} . In other words, a higher level of hydrophobicity is obtained upon increasing the air under the contact line. According to Eq. (1), for the optimum sample (MEK6) having θ_{SH} and θ_0 equal to 162 and 104°, respectively, f_2 value is calculated to be 0.9354 implying that air occupies about 93.5% of the contact areas when the rough hydrophobic polymer surface contacts with the water droplet.

3.1.2. Singular use of silica nanoparticles

Table 1 shows the wettability results for the coatings with different nano-silica contents (Si50, Si100 and Si150). As is evident, only addition of 100 mg nano-silica into PP solution causes the water contact angle value to rise above 150°. A substantial difference can also be seen in the sliding angle values. Therefore, it is clear that Si50 follows the Wenzel's regime while the Si100 follows the Cassie's regime. It is worth noting that the surface of sample containing 150 mg (Si150) was unstable and could be easily rubbed off, and thus, wettability results were not measured for this sample.

Table 1

Wettability results regarding singular use of silica and combined use of silica and MEK.

Samples	Si50	Si100	Si150	MEK2Si10	MEK2Si20	MEK2Si30
Water contact angle (°)	132 ± 4	155 ± 2	–	147 ± 4	166 ± 1	165 ± 3
^a Sliding angle (°)	60	6	–	20	1	1

^a The angle at which the water droplet starts to roll off the surface upon tilting.**Table 2**XPS atomic contents (at.%) for the coatings containing 50 and 100 mg SiO₂.

Atom	Si50 (at.%)	Si100 (at.%)
C	60.3	36.2
O	26	41.7
Si	13.4	21.6

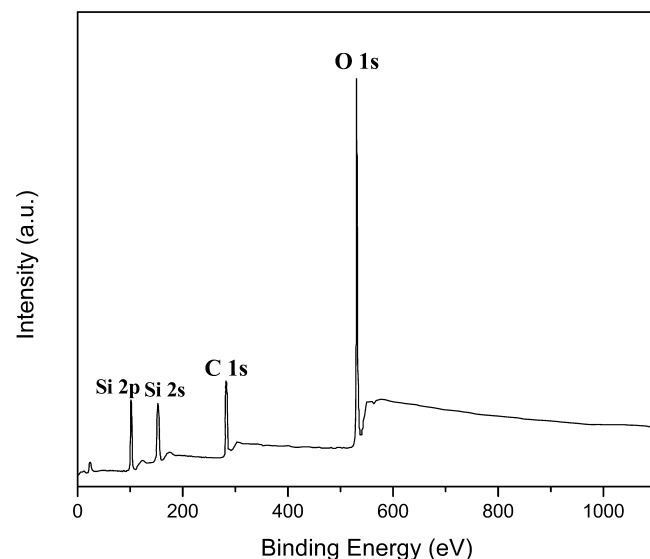
Fig. 2c and d shows SEM images for the samples containing 50 and 100 mg SiO₂ (Si50 and Si100). Upon comparing those two images, one could observe that the silica nanoparticles aggregated densely when the content of SiO₂ increased to 100 mg leading to a highly packed morphology. In the case of Si50 sample, silica nanoparticles cannot cover the PP surface completely which leads to exposure of bare and utterly smooth PP surface to the water droplet resulting in a much higher sliding angle value.

It could be postulated that the observed wetting behavior of the as-prepared surface is a result of different surface compositions. Therefore, XPS was used to investigate the surface compositions of the samples containing 50 and 100 mg silica nanoparticles. The obtained elemental concentrations of Si, O, and C demonstrated that SiO₂ and PS coexisted on the as-prepared surface as reported in Table 2. The ratio of Si, O, and C further revealed the change of surface composition when the amount of nanosilica increased to 100 mg. As can be seen in Table 2, the ratio of Si/O/C is 13.4/26/60.3% when the sample possesses 50 mg nanosilica. On the other hand, upon increasing the amount of nanosilica to 100 mg, the ratio of Si/O/C changes to 21.6/41.7/36.2. It is seen that the atomic concentration of Si increased from 13.4 to 21.6% indicating that nanosilica aggregated gradually on the surface as the amount of nanosilica increased.

This can explain the wetting transition between hydrophobicity and superhydrophobicity for the coatings. In the case of using low nanosilica content (50 mg), PP macromolecules are more likely to aggregate on the top surface, and therefore, the coating is hydrophobic once the silica content is low. Conversely, when the nanosilica content is increased, the hydrophobic material (SiO₂) becomes the main component of the coating, and thus, the as-prepared surface shows superhydrophobicity. Such XPS results utterly corroborate the made claims based on SEM observations.

3.1.3. Combined use of nanoparticles and non-solvent

To evaluate concurrent effect of non-solvent and nanoparticles on wettability behavior of the coatings, proper combinations of MEK and nanosilica were added into PP/xylene solution. The prepared coatings exhibited different appearances and mechanical stabilities. According to Table 1, it is seen that the sample containing 2 ml MEK as non-solvent and 10 mg nanosilica (MEK2Si10) exhibits a water contact angle of 147° but the sliding angle is rather high, and thus, cannot be regarded as a superhydrophobic surface. Whereas, upon rising the nanosilica content to 20 mg (MEK2Si20) a notable enhancement and reduction was detected for water contact angle value (166°) and sliding angle (1°), respectively; hence, such composition resulted in a superhydrophobic coating. It is interesting to note that further addition of nanosilica content to 30 mg, at the fixed non-solvent content, has resulted in a similar wettability behavior as in the case of MEK2Si20; however, many cracks were macroscopically detected on the coating surface leading to

**Fig. 3.** XPS wide-scan spectrum for the optimum composition (MEK2Si20).

a slightly poorer adhesion to the substrate. In conclusion, 20 mg nanosilica at fixed MEK content (2 mL) can be considered as the optimal composition.

Fig. 2e and f shows surface morphologies for samples containing 10 and 20 mg nanosilica made from a fixed amount of non-solvent (MEK2Si10 and MEK2Si20), respectively. It is evident that both surfaces are covered by ultra-fine silica nanoparticles, and are rather similar to each other; however, there exist some differences. In the case of MEK2Si10, the surface has not been covered by nanospheres completely (marked in Fig. 2e), and as a result, the water contact angle for this sample is lower than that of other samples in this category (see Table 1). A closer look at Fig. 2f and g reveals that nanospheres covered the surface completely with an average diameter of 150 nm, which leads to an ordered type of surface roughness. Such result could be taken into consideration from the geometrical viewpoint, that is, entrapment of more air pockets because of the nano-sized surface features as compared to the micro-sized ones based on the model presented in Ref. [2]. In fact, surface homogeneity is enhanced because of the increased water/air contact area. Therefore, the surface heterogeneity, which is indicative of water contact angle hysteresis, is greatly diminished. Moreover, it is worthy to note that addition of higher silica content changed the surface morphology insignificantly; however, several cracks were macroscopically detected on the coating surface indicative of higher silica content on the top layer of the coating.

In order to determine the surface composition of the optimum sample made from both nanosilica and MEK, wide-scan XPS spectrum for this sample was obtained (Fig. 3). It was found that a great deal of silica nanoparticles was present on the surface explaining its ultra-high water repellency behavior. The surface elemental composition was calculated from the obtained peaks and the ratio of Si/O/C is 24.1/46.8/28.8. Such result is well correlated with the obtained morphology in Fig. 2f and g demonstrating a high level of packing for nanospheres which can only be detected in the case of sample containing 100 mg nanosilica in the absence of

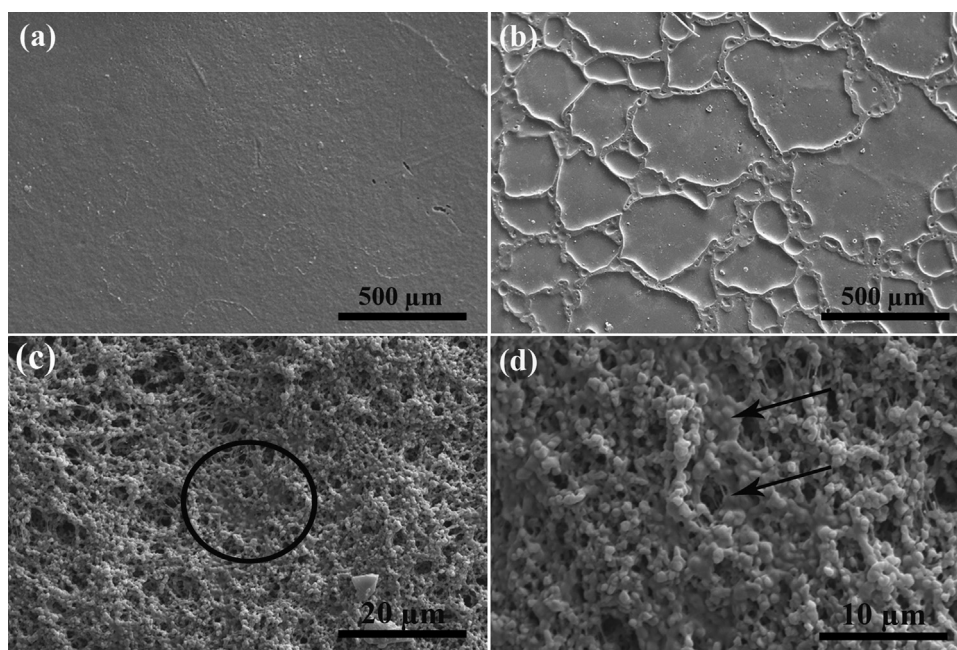


Fig. 4. SEM images for (a) uncultured pure PP, (b) cell cultured pure PP, (c) cell cultured MEK6 sample, and (d) higher magnification image of MEK6 sample.

non-solvent. Based on the above result and Table 2, the interesting observation is that although the silica content is much lower for MEK2Si20 as compared to Si150 (20 mg versus 150 mg), the percentage of Si atoms is higher for MEK2Si20 implying the efficiency of the employed method in collecting silica nanoparticles into the top layer of the coating. Therefore, it could be inferred that the combined use of nanoparticle and non-solvent caused the required amount of nanoparticle for superhydrophobicity to be diminished substantially.

3.2. Cell adhesion

In this study, 4T1 mouse mammary tumor cells were used to evaluate the effects of surface wettability and, in particular, topography on the behavior of cultured cells. First, it is worth considering how the cells adhered onto the flat PP surfaces which were made through melting between two glass slides. To this end, scanning electron microscopy (SEM) analysis was utilized since this approach is able to distinguish substrate-dependent differences in the shape of individual cells in high-density cultures.

Fig. 4a and b shows the morphology of pure PP before and after cell culture, respectively. As anticipated, a number of cells were adhered onto the control smooth PP surface showing a moderate level of spreading on the substrate, which could be due to the intrinsically low surface energy of PP. Fig. 4c and d displays that the cells adhered less onto the superhydrophobic MEK6 surface as compared to the control smooth PP surface (Fig. 4b). It should be noted that a large SEM scan was required for observing such detectable adhered cells. Moreover, the shape of the cells was changed and became much narrower. This phenomenon could be due to the fact that, according to the Cassie and Baxter model, the medium suspension cannot be completely contacted with the entire surface.

In order to determine the effect of surface topography, samples made from a combined use of nanoparticle and non-solvent, which have exhibited an ultra-fine morphology, were also subjected to cell culture. As can be seen in Fig. 5a, according to a low magnification view for MEK2Si10, no detectable adhered cells could be observed on such a large surface area of the coating. The same result was observed for MEK2Si20, and the corresponding SEM image is shown

in Fig. 5c. However, both surfaces exhibit some inhomogeneity and asperities, which were highly magnified in order to detect any adhered cells. Fig. 5b and d clearly illustrates that cells preferentially attached to the crevices between the surface asperities while the other regions of the topographical features were cell-free. Such results indicated that the cell adhesion process was severely inhibited once the nano-scale roughness is particularly small.

Another observation based on Fig. 5b and d is the extended morphology of the cells adhered on the superhydrophobic PP surface via cytoplasmic projections, which has been suggested to be the result of significantly suppressed contact between the cells and the entire material surface [27].

For a better evaluation, the cytocompatibility of samples was determined via quantitative MTT assay at three and five days after the seeding. Based on Fig. 6, one could observe that cells seeded in pure PP sample remained rather viable (78% after three days and 73% after 5 days) throughout the culture period. Apart from that, a notable reduction of cytocompatibility was observed for MEK6 sample which had exhibited a micron-sized structure; however, such observed reduction is not in conformity with the results of Alves et al. [27] who reported a prevention in cell adhesion by superhydrophobic coatings based on poly(L-lactic acid).

However, in the case of MEK2Si20 and MEK2Si10 with nano-scale topographical features, an interesting observation is that the density of viable cells over those surfaces was remarkably lower than that over both pure PP and superhydrophobic MEK6 samples. Such result indicates a great loss in the viability of the adhered cells in the seeding step which could be most probably due to their inability to spread along such water repellent coating. Therefore, the role of surface topography should be more emphasized as compared to the role of superhydrophobicity. It is true that an extremely water repellent surface could lead to a non-proliferative behavior; however, it is shown in this study that the surface topography plays an important role in the cell adhesion behavior as well. Our results imply that 4T1 cells, used in this study, respond differently to surfaces with various micro/nano-scale topographical features evidently demonstrating a certain level of inhibition in cell adhesion once the surface is only comprised of nano-scale roughness.

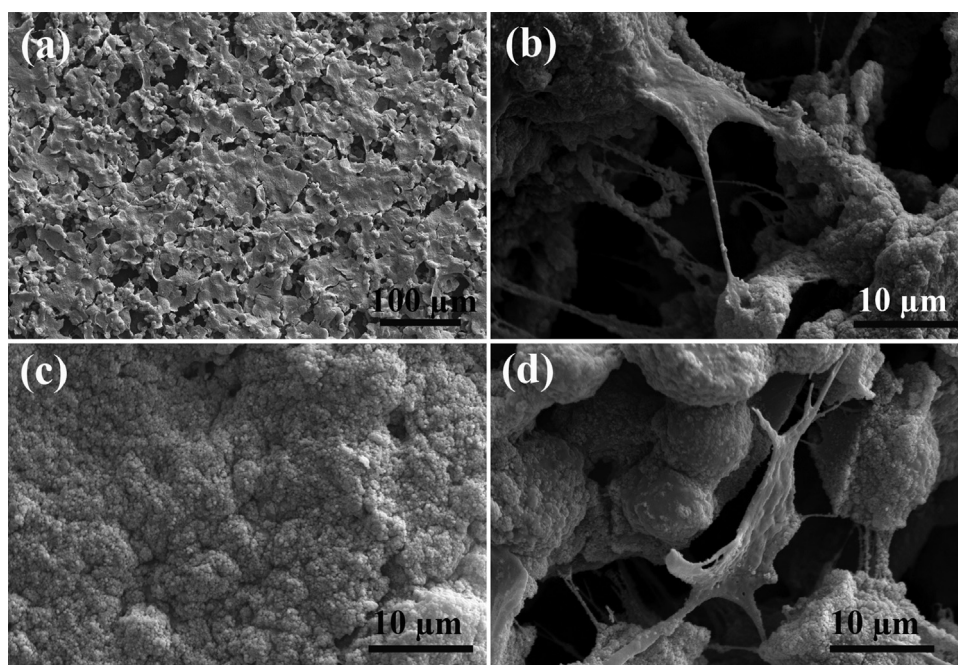


Fig. 5. SEM images at different magnifications for (a,b) cell cultured MEK2Si10, and (c,d) cell cultured MEK2Si20.

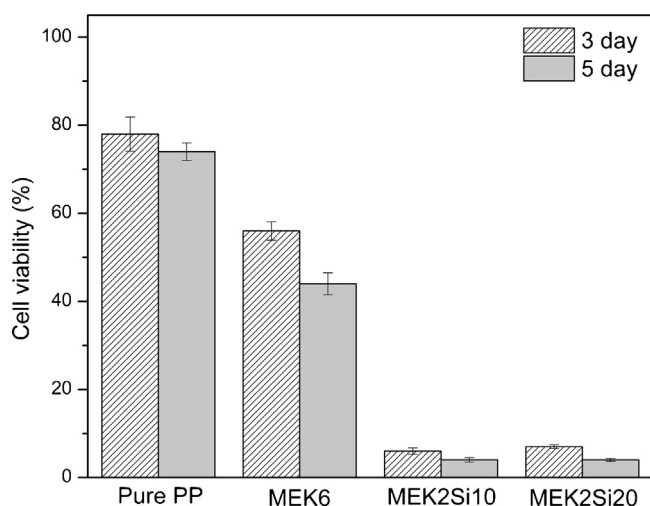


Fig. 6. Cell viability for different samples at 3 and 5 days of postseeding.

4. Conclusions

A novel method including combined use of silica as nanoparticles and MEK as non-solvent was employed in this study to prepare superhydrophobic PP-based coatings with different surface topographical features. It was systematically proved that individual use of nanoparticles and non-solvent could lead to superhydrophobicity but at the price of adding high contents. From the practical viewpoint, the prepared coatings exhibited a rather homogeneous appearance and proper adhesion to the substrates, which have not been observed in the case of individual uses of nanoparticles and non-solvent. Therefore, different surface topographical features were developed using both the proposed method and the method presented by the Erbil et al. The results demonstrated that under the employed conditions (surface composition, surface morphology, and cell type etc.), the cellular response is highly inhibited once the surface structuring is pushed toward nanometric dimension as compared to the flat PP surface. Although superhydrophobicity

of the prepared surfaces is believed to play an important role in inducing such inhibition, it was found that the surface micro/nano structure or, in other words, dimensions of topographical features is also another vital parameter that has to be considered during the developing process. In our future studies, different surface chemistries and topographical features will be investigated to further elucidate the specific cellular responses.

Acknowledgment

Partial financial support from the Iranian Nanotechnology Initiative is gratefully appreciated.

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